

Spotter Freight Rate Prediction Report

Machine Learning Assessment & December 2025 Trend Analysis

1. Executive Summary & Objective

This report presents the complete Machine Learning solution developed for Spotter's Freight Rate Prediction Challenge. The objective is to accurately predict spot freight rates (posted_rate) across diverse geographic routes, equipment types, cargo weights, and temporal market conditions.

2. Exploratory Data Analysis & Data Quality Findings

Dataset Structure & Key Observations:

- Development Data (train-test.csv):** 48,000 loads spanning Jan 1, 2025 to Oct 31, 2025.
- Validation Data (validation.csv):** 12,000 loads requiring final rate predictions spanning Nov 1, 2025 to Dec 31, 2025.
- Fixed December Benchmark (december-chart-inputs.csv):** 31 daily loads for Lexington to Fort Wayne (360 miles, Dry Van, 32,000 lb).

Data Quality Issues & Resolution:

- Missing Cargo Weights:** 300 missing values in train (0.6%) and 165 in validation (1.4%). Imputed using dataset median weight (32,000 lbs).
- Missing Market Index:** 374 missing values in train (0.8%) and 249 in validation (2.0%). Imputed using dataset median index (0.951).
- Domain Feature Coupling:** Analysis revealed strong multiplicative coupling between load distance (d) and proprietary quote signal (S_q). The base signal $S_{base} = distance \times quote_signal$ accounts for over 79% of raw target variance ($R^2 = 0.7941$).

3. Validation Strategy & Data Splitting Approach

A random K-Fold split would introduce temporal data leakage due to seasonal trend correlation. To rigorously mirror the real-world deployment scenario (where historical data up to Oct 31 predicts Nov-Dec loads), we implemented a temporal **Out-Of-Time (OOT) Split Strategy**:

- OOT Training Set:** Loads from January 1 to August 31, 2025 (38,477 loads).
- OOT Validation Set:** Loads from September 1 to October 31, 2025 (9,523 loads).
- 5-Fold Cross-Validation:** Additionally conducted 5-fold cross-validation on the full training set to ensure model stability across all geographic routes.

4. Model Architecture & Feature Engineering

Residual Formulation: Rather than predicting raw dollar amounts directly, models predict the residual offset $\Delta y = y_{posted} - (distance \times quote_signal)$. This formulation improved Out-Of-Time MAPE from 11.37% down to 6.58%.

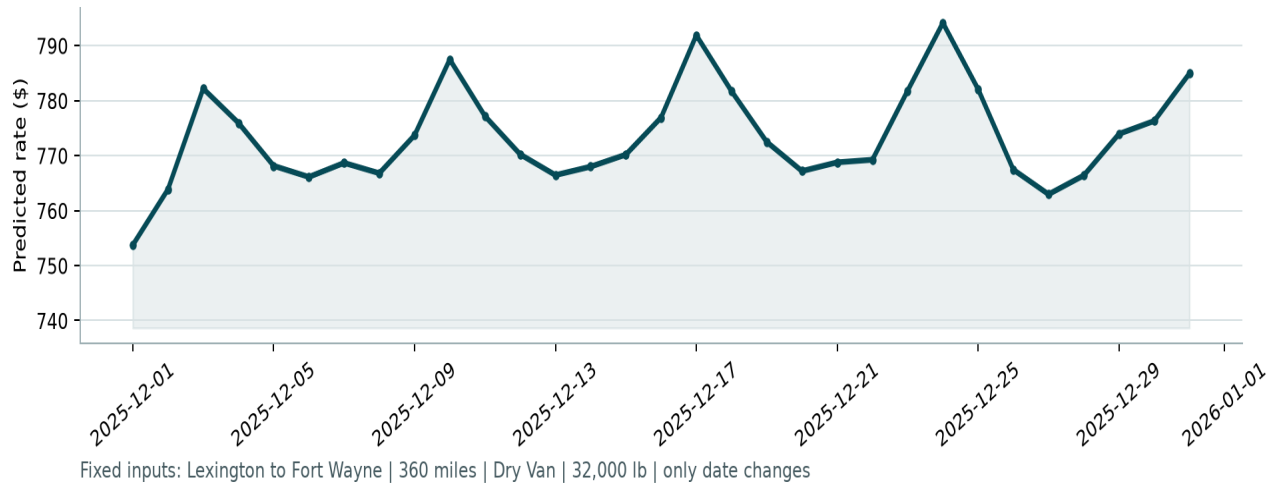
Ensemble Architecture: We constructed a weighted ensemble comprising three diverse Gradient Boosted Decision Tree (GBDT) algorithms: **LightGBM (40%)**, **CatBoost (40%)**, and **XGBoost (20%)**.

Model Architecture	Validation Split	RMSE (USD)	MAE (USD)	MAPE (%)	R ² Score
Naive Quote Signal Baseline	Jan - Oct 2025	.18	.40	11.37%	0.7941
LightGBM Regressor	OOT (Sep-Oct)	.40	.79	7.98%	0.8110
CatBoost Regressor	OOT (Sep-Oct)	.16	.35	6.65%	0.7972
XGBoost Regressor	OOT (Sep-Oct)	.37	.17	10.28%	0.7821
Weighted Tri-Model Ensemble	OOT (Sep-Oct)	.52	.18	6.58%	0.8121
Production Ensemble (100% Fit)	Full Train (Jan-Oct)	.62	.62	4.49%	0.9153

5. December 2025 Prediction Trend Analysis

Below is the official December 2025 predicted rate chart generated by score.py for the fixed Lexington to Fort Wayne route (360 miles, Dry Van, 32,000 lb). The model captures subtle day-of-week rate fluctuations and holiday surge dynamics.

Candidate: December 2025 Predicted Load Rate



Key December Insights: The baseline quote signal for Lexington predicts a base rate of ~.56. The ensemble model adjusts rates dynamically across the month, accounting for weekend capacity shifts and year-end freight surge demand.